

Nikhilesh Vijay Chirutala

Software Engineer | Python | AI/ML | AWS

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SUMMARY

Software Engineer with 2.8+ years of experience building production-grade Python applications, backend services, REST APIs, AI-enabled applications, and cloud-based systems across contact center, CRM, trading, and business platforms. Strong hands-on experience with Python, FastAPI, Django, Flask, AWS, SQL, MongoDB, REST APIs, CI/CD, Git, and production troubleshooting. Experienced in developing scalable backend services, distributed service integrations, cloud storage workflows, authentication and authorization, monitoring, and reliable production deployments across multiple environments. Hands-on experience building RAG-based AI solutions and integrating LLM capabilities for contextual information retrieval and application workflows. Proven ability to design, develop, test, troubleshoot, and continuously improve secure and reliable software components in production environments.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Software Engineering: Object-Oriented Programming, Data Structures, Algorithms, REST API Design, API Development, Backend Development, Microservices, Distributed Services, Software Debugging

Backend & Frameworks: FastAPI, Django, Django REST Framework, Flask, REST APIs, Backend Automation, Business Logic, API Integration

AI & Machine Learning: AI/ML Applications, RAG, LLM Integration, Vector Search, Prompt Engineering, Contextual Information Retrieval

Cloud & AWS: AWS, Amazon S3, Cloud Application Development, Cloud Deployments, Production Cloud Services, Vercel

Databases: MySQL, MongoDB, PostgreSQL, Neon DB, SQL, Database Design

DevOps & SDLC: Git, GitHub, GitHub Actions, CI/CD, Linux, Feature Flags, Staged Rollouts, Production Deployment, Monitoring, Automation

Security: Authentication, Authorization, RBAC, ACL, JWT, Secure API Development, Access Control

Observability & Reliability: Datadog, StatsD, Sentry, Production Monitoring, Error Tracking, Troubleshooting, Failure Recovery, Retry Strategies, Validation, Incident Resolution

APIs & Integrations: REST APIs, Twilio, GitHub API, Postman, Third-Party Integrations, Service-to-Service Integration

Frontend: React.js, Next.js, Angular, Tailwind CSS, Bootstrap, Jinja2

Testing & Tools: Unit Testing, Postman, Jest, Git, GitHub, Debugging, API Testing, Validation

WORK EXPERIENCE

Python Software Engineer • CA One Tech Cloud Inc.

Apr 2026 – Present | Bengaluru, India

Client: Convoso — Cloud Contact Center Platform

Python, FastAPI, AWS, Amazon S3, MySQL, Datadog, StatsD, Sentry

- Develop and maintain production-grade Python and FastAPI services for a high-volume cloud contact center platform, supporting call-recording ETL, data processing, storage workflows, service integrations, and production operations.
- Migrated a legacy Perl-based call-recording workflow to a long-running Python service deployed across 40 production clusters, implementing feature flags, staged rollouts, validation, rollback, monitoring, and production troubleshooting.
- Designed and implemented Python service integrations with a Call Intelligence microservice, handling API communication, eligibility checks, account and campaign controls, daily processing limits, retries, response validation, and result persistence.
- Developed reliable validation and recovery workflows for call-recording data, handling incomplete processing, failed operations, inconsistent storage states, and downstream processing dependencies across production environments.
- Supported secure and scalable production deployments of Python services across multiple clusters, coordinating feature-flag configuration, validation, monitoring, troubleshooting, and staged rollout activities.
- Built AWS S3 audit and recovery workflows for recording storage, implementing deterministic object handling and idempotent re-upload operations for incomplete or failed storage migrations.
- Implemented production observability using Datadog, StatsD, dashboards, and Sentry to monitor service health, track application errors, diagnose processing failures, and support incident resolution.
- Investigated and resolved production issues across Python services, REST APIs, databases, and AWS cloud storage, identifying root causes and implementing fixes to maintain reliable and scalable processing workflows.

Software Engineer • Mantra Technologies

Feb 2024 – Mar 2026 | Hyderabad, India

- Worked across four production applications, delivering Python backend services, REST APIs, frontend features, third-party integrations, authentication, and application workflows across the software development lifecycle.
- Designed and developed backend modules, REST APIs, and business logic using Python, FastAPI, and MongoDB for production application workflows.
- Developed AI-enabled application capabilities, including an AI-powered trading platform and Retrieval-Augmented Generation (RAG) chatbot for contextual information retrieval.

- Built frontend interfaces using React.js, Next.js, Angular, JavaScript, and TypeScript, integrating application workflows with backend APIs and services.
- Implemented authentication, RBAC, and ACL-based authorization to enforce secure access controls across application resources and user workflows.
- Integrated third-party services including Twilio and GitHub API and supported application deployments across AWS and Vercel environments.
- Maintained GitHub Actions CI/CD workflows and contributed to continuous integration, testing, debugging, bug fixes, production support, and reliable application releases.

Key Projects:

AI Bull — AI-Powered Trading & AI/ML Platform

React.js, Next.js, FastAPI, Python, MongoDB, RAG, LLM

- Developed an AI-powered trading and investment analysis platform for stock and mutual fund analysis, screening, back-testing, and data-driven application workflows.
- Implemented a Retrieval-Augmented Generation (RAG) based chatbot that retrieves relevant financial and application-specific information to generate contextual LLM-powered responses.
- Integrated LLM capabilities into application workflows, enabling conversational access to relevant financial information and contextual application data.
- Built FastAPI and Python backend APIs and business logic supporting data processing, AI-enabled workflows, and application services.
- Developed back-testing engines and configurable screening and filtering capabilities for stocks and mutual funds.

CRM.io — AI-Driven CRM & Communication Platform

FastAPI, Python, Jinja2, MongoDB, ACL, Twilio, REST APIs

- Developed production CRM modules for Contacts, Tasks, communication workflows, and telephony operations using FastAPI and Python.
- Designed and implemented REST APIs and backend business logic for CRM workflows and application data operations.
- Implemented RBAC and ACL authorization to enforce secure, permission-based access to CRM resources and workflows.
- Integrated Twilio for telephony and communication functionality and developed server-rendered interfaces using Jinja2.
- Tested APIs and end-to-end workflows using Postman and resolved defects across CRM functionality and production workflows.

GharPe — Real Estate Management Platform

React.js, FastAPI, Python, MongoDB, REST APIs

- Developed the React.js-based administration system for managing real estate projects and controlling their visibility and availability across the platform.
- Built project management workflows allowing administrators to create, update, configure, and manage projects displayed across the platform.
- Developed lead management functionality to track and manage customer inquiries generated through the platform.
- Built builder management features for maintaining builder information and associating builders with their respective projects.
- Developed and integrated FastAPI/Python REST APIs supporting administrative workflows, project configuration, validation, and data operations.

PERSONAL PROJECTS

JS Studio — Online JavaScript Playground

Next.js, GitHub API, Vercel

- Developed an online JavaScript playground for writing and executing JavaScript through a web interface.
- Integrated GitHub APIs and deployed the application with Next.js and Vercel using a production-oriented application architecture.

AnyScrum — Project Management Platform

Next.js, Neon DB, Clerk, Vercel

- Developed a project management platform supporting projects, tasks, and team workflows.
- Implemented authentication and user management with Clerk and persistent application data with Neon DB.

Guider — Interactive Learning Portal

Angular 19, TypeScript, SSR, Vercel

- Built an interactive learning platform using Angular 19, TypeScript, and server-side rendering with reusable frontend components.
- Deployed the application on Vercel with a production-ready frontend architecture.

Enterprise REST API Platform — Django and Flask Backend Application

Python, Django, Django REST Framework, Flask, PostgreSQL, REST APIs

- Built modular REST APIs using Django, Django REST Framework, and Flask for authentication, business logic, and data-driven workflows.
- Integrated PostgreSQL with CRUD operations, database models, relationships, validation, and API error handling.
- Implemented JWT authentication and role-based access control with API testing and validation using Postman.

EDUCATION

B.Tech — Electrical & Electronics Engineering

2019 – 2023 | GPA: 7.73 / 10

Srinivasa Ramanujan Institute of Technology, Anantapur